input

```
Enter total number of frames: 7
Enter window size: 3

Sliding Window Protocol Simulation

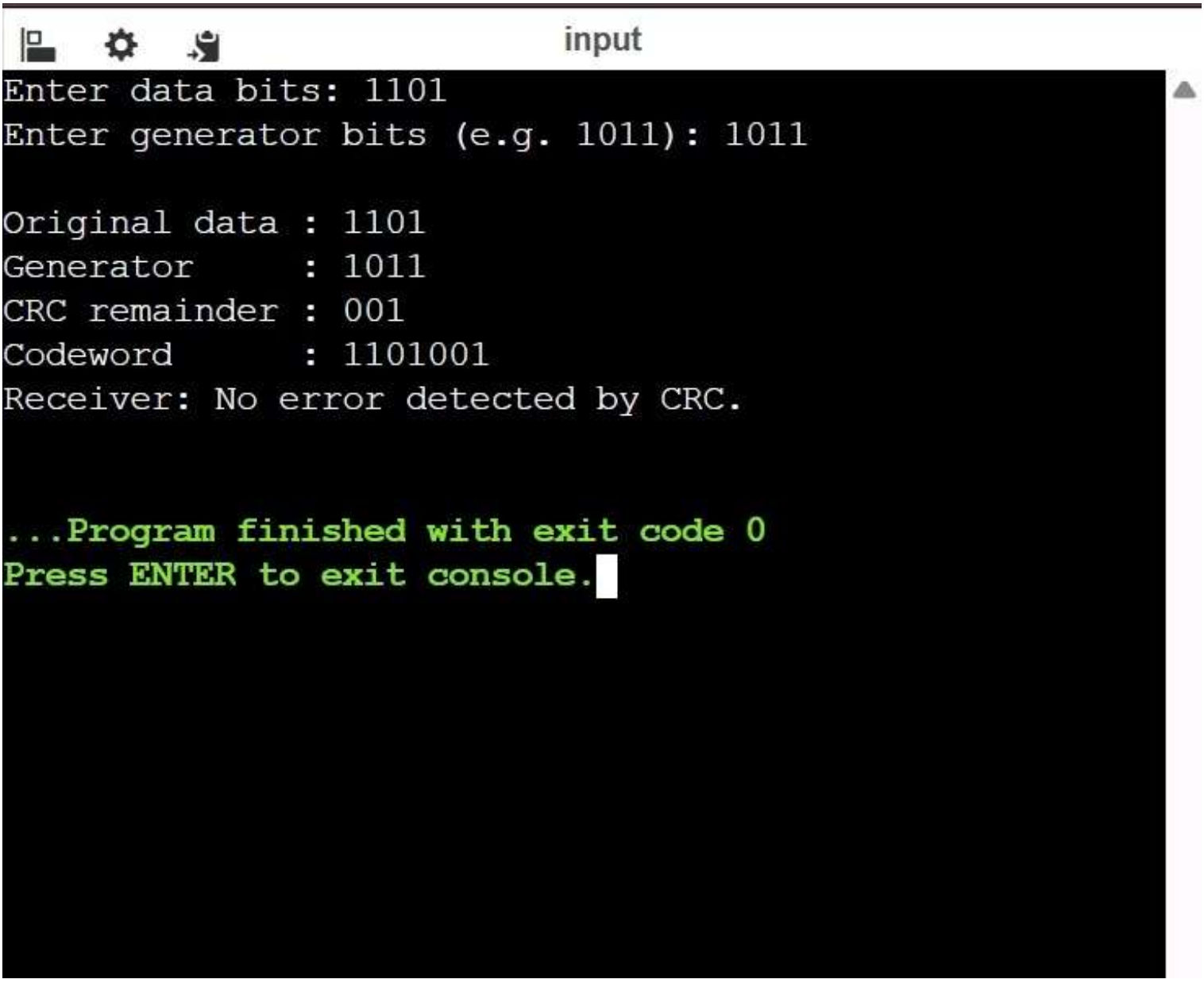
Sender window: frames 0 to 2
Sender: Sending frame 0
Sender: Sending frame 1
Sender: Sending frame 2
Receiver: Frame 0 received
Receiver: ACK 0 sent
Receiver: Frame 1 received
Receiver: ACK 1 sent
Receiver: Frame 2 received
Receiver: ACK 2 sent
Sender: Window moved forward

Sender window: frames 3 to 5
Sender: Sending frame 3
Sender: Sending frame 4
Sender: Sending frame 5
Receiver: Frame 3 received
Receiver: ACK 3 sent
Receiver: Frame 4 received
Receiver: ACK 4 sent
Receiver: Frame 5 received
Receiver: ACK 5 sent
Sender: Window moved forward

Sender window: frames 6 to 6
Sender: Sending frame 6
Receiver: Frame 6 received
Receiver: ACK 6 sent
Sender: Window moved forward
```

```
Enter data bits (0s and 1s only): 111110111111
Original data : 111110111111
Stuffed frame : 0111111011111001111110101111110


...Program finished with exit code 0
Press ENTER to exit console.
```


input
Enter data bits: 1101
Enter generator bits (e.g. 1011): 1011
Original data : 1101
Generator : 1011
CRC remainder : 001
Codeword : 1101001
Receiver: No error detected by CRC.
...Program finished with exit code 0
Press ENTER to exit console.